

Title: *Avian Migration: A Study of Migratory Birds Visiting Various Habitats in Maharashtra*

Aim: To gather information on migratory birds visiting various habitats in Maharashtra, document their taxonomic classification, analyze their ecological importance, and arrive at a conclusion based on the analysis of data.

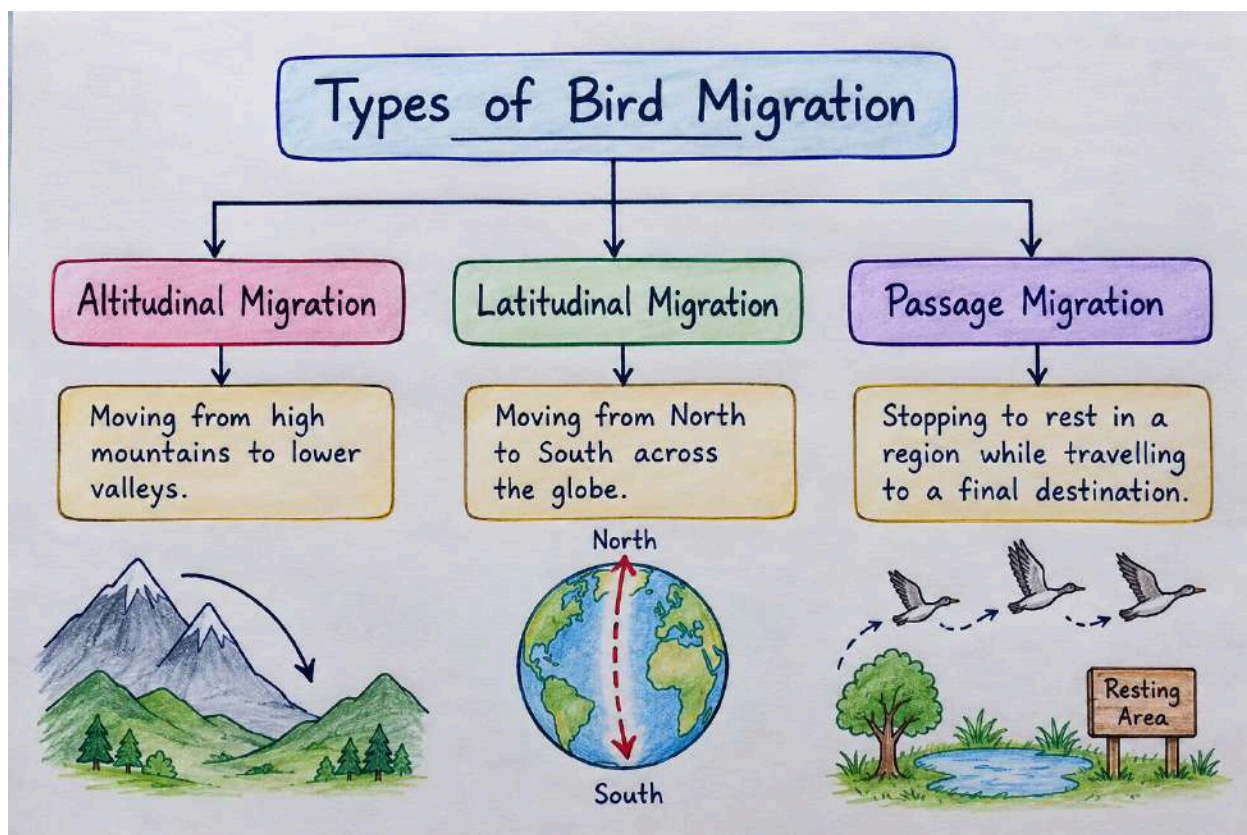
Project Writing

1. Name of the student and concerned details

- **Name:** Rudra Raju Varankar
- **Standard:** 12th (Science)
- **Subject:** Biology
- **Project Topic:** *14 (Gather the information of Migratory birds visiting various habitats in Maharashtra)*
- **Academic Year:** 2026-2027

2. Acknowledgements: I would like to express my sincere gratitude to my Biology teacher for their invaluable guidance in planning the approach and methodology for this project. Their continuous support helped me understand complex ecological concepts clearly. I am also deeply thankful to our Principal for providing the necessary academic environment. Lastly, I extend my heartfelt thanks to my parents and classmates for their encouragement and assistance in compiling this data.

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3. Table of contents

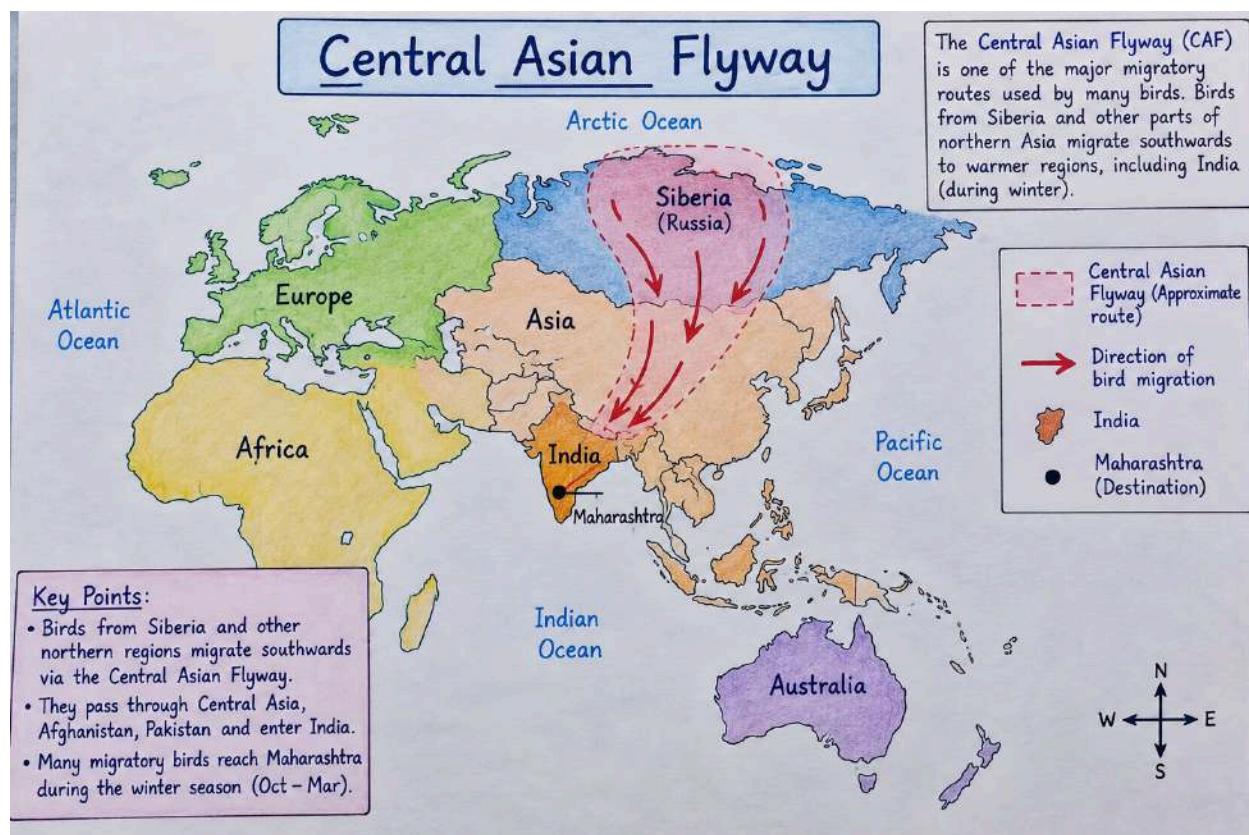
1. Title and Aim
2. Name of the student and concerned details
3. Acknowledgements
4. Table of contents
5. Actual Report a. Introduction of the topic and need for the project b. Description of method/s used c. Data collected (Winter and Passage Migrants) d. Analysis of collected data e. Conclusion
6. Bibliography

4. Actual Report

a. Introduction of the topic and need for the project Introduction to Avian Migration:

Migration is a remarkable biological phenomenon where animals undertake long-distance journeys triggered by seasonal changes, food availability, and breeding cycles. Maharashtra, with its diverse geography comprising the long Konkan coastline, the towering Western Ghats, and the expansive Deccan Plateau wetlands, serves as a crucial wintering ground and transit hub for millions of migratory birds utilizing the Central Asian Flyway.

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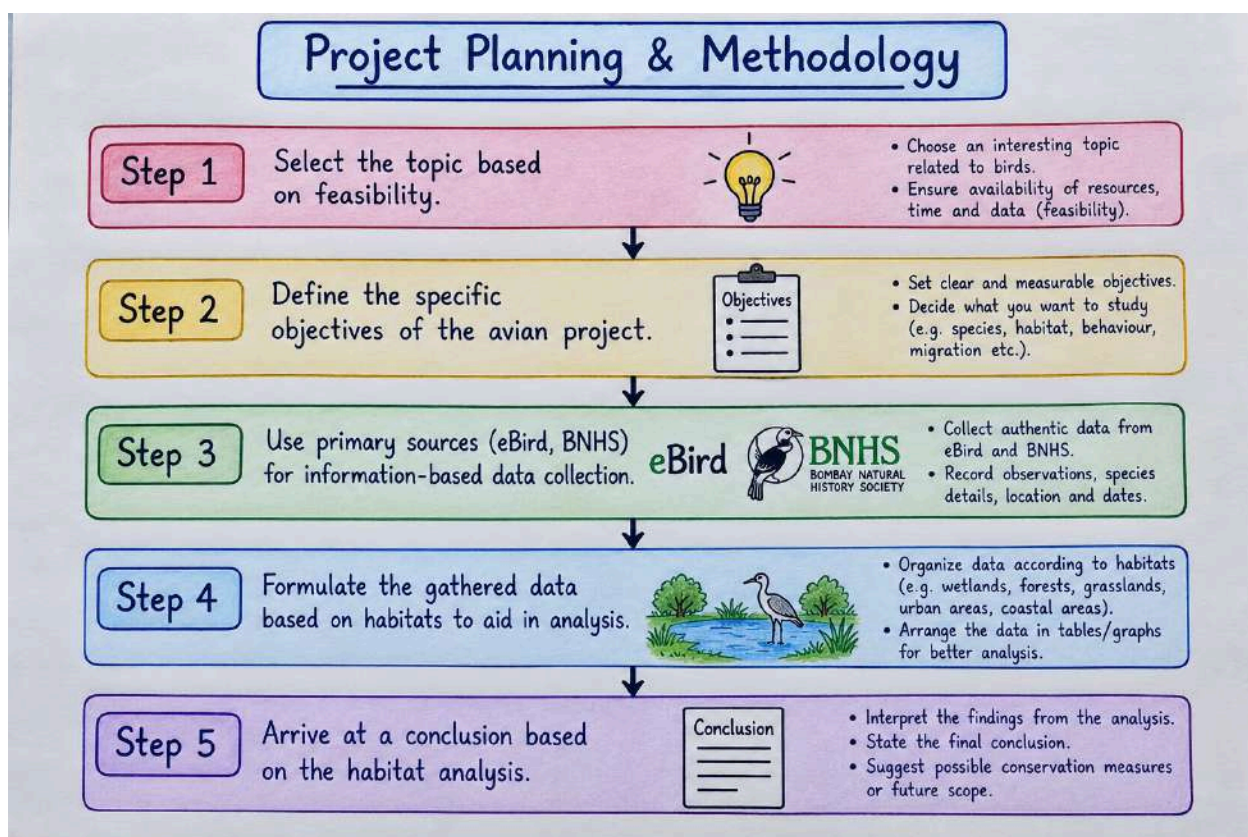


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Every winter, hundreds of species travel thousands of kilometers from the freezing climates of Siberia, Central Asia, and Europe to the warm, resource-rich habitats of Maharashtra. These avian visitors utilize various habitats, including coastal mudflats (like Sewri in Mumbai), inland freshwater lakes (like Jayakwadi and Ujani dams), and open grasslands.

Need for the Project: Migratory birds are exceptional bio-indicators of environmental health. However, their populations are declining globally due to the destruction of wetlands, climate change, and coastal pollution. This project is highly necessary to systematically document the specific migratory species visiting Maharashtra. By understanding their habitat requirements and identifying the unique threats they face during their journey, we can advocate for the protection of these vital ecological wintering grounds.

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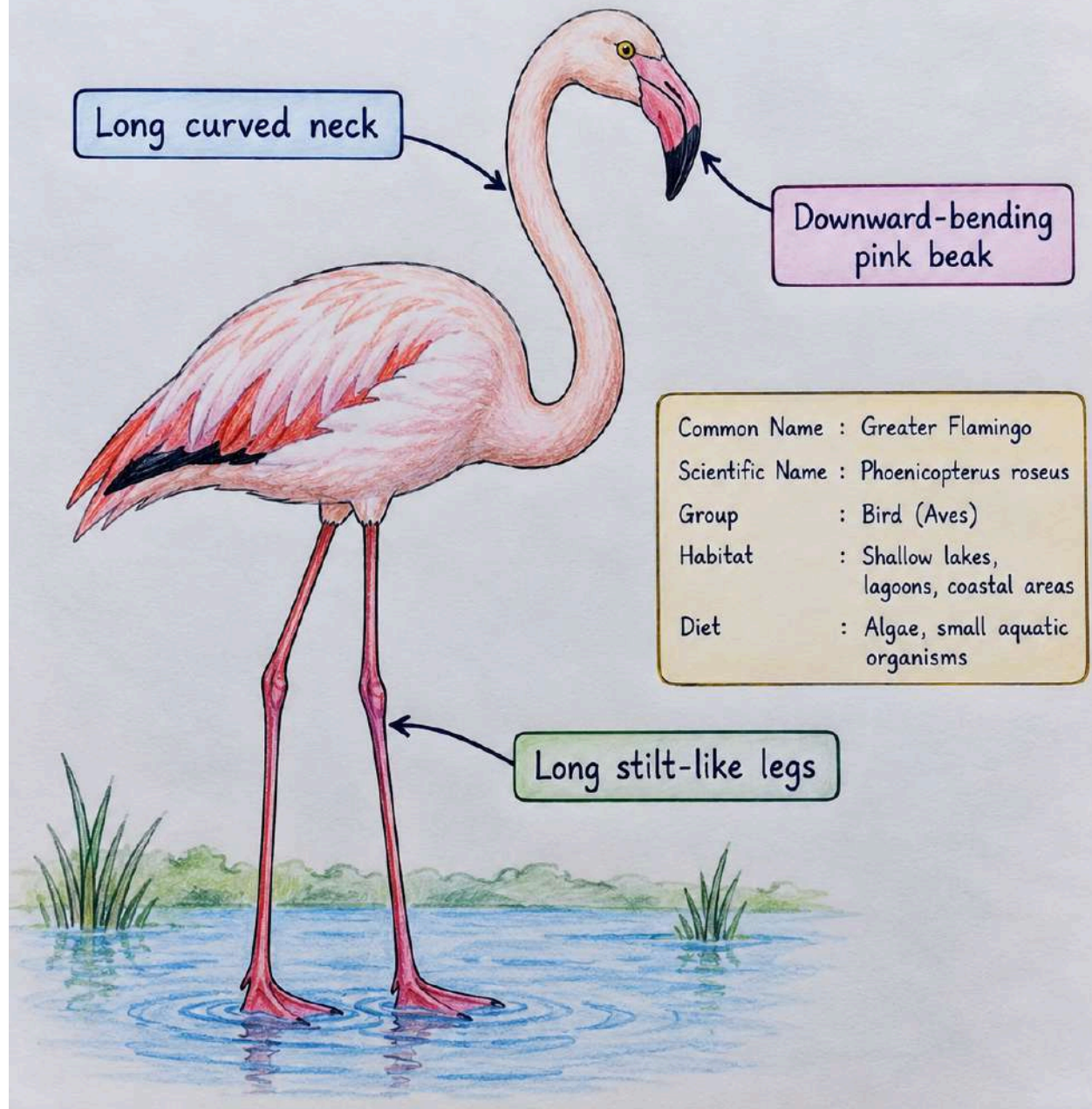
b. Description of method/s used As per the strict project guidelines for this topic, the approach and methodology rely on primary sources for an information-based project. Direct experimentation for collection is not feasible due to the protected status of these birds and their seasonal availability.

- Primary Sources:** Secondary data was meticulously collected from highly reliable, peer-reviewed ornithological sources. Information on species distribution was referenced from the Bombay Natural History Society (BNHS) and global avian databases like eBird.
- Data Formulation:** Specific habitat details and localized distribution within Maharashtra were gathered using various techniques so as to aid in analysis.
- Data Organization:** The information was systematically categorized into specific bird species, detailing their taxonomic classification, wintering habitat in Maharashtra, unique morphological characteristics, and the primary threats they face.

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Greater Flamingo

(*Phoenicopterus roseus*)



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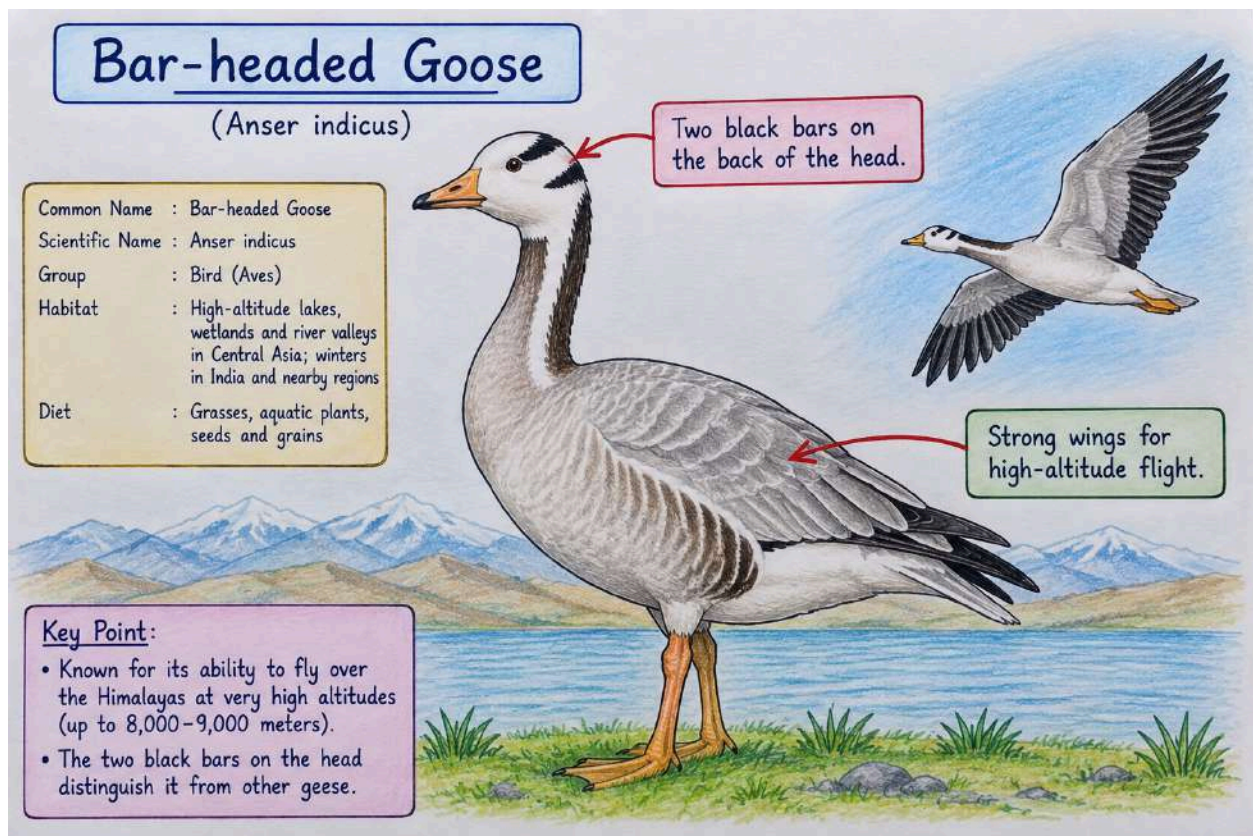
c. Data collected

Part I: Wetland and Coastal Migrants

1. Greater Flamingo

- **Taxonomic Classification:**
 - Kingdom: Animalia
 - Phylum: Chordata
 - Class: Aves
 - Order: Phoenicopteriformes
 - Scientific Name: *Phoenicopiterus roseus*
- **Migratory Status:** Winter Migrant (from Rann of Kutch and Central Asia).
- **Habitat in Maharashtra:** Coastal mudflats (Sewri, Mumbai) and inland shallow wetlands (Ujani Dam, Bhigwan).
- **Characteristics:** A highly social, tall wading bird known for its distinctive pinkish-white plumage, long curved neck, and specialized downward-bending beak used for filter-feeding on algae and shrimp.
- **Primary Threats:** Coastal reclamation, industrial pollution in mudflats, and disturbances caused by unregulated tourism and boating in shallow wetlands.

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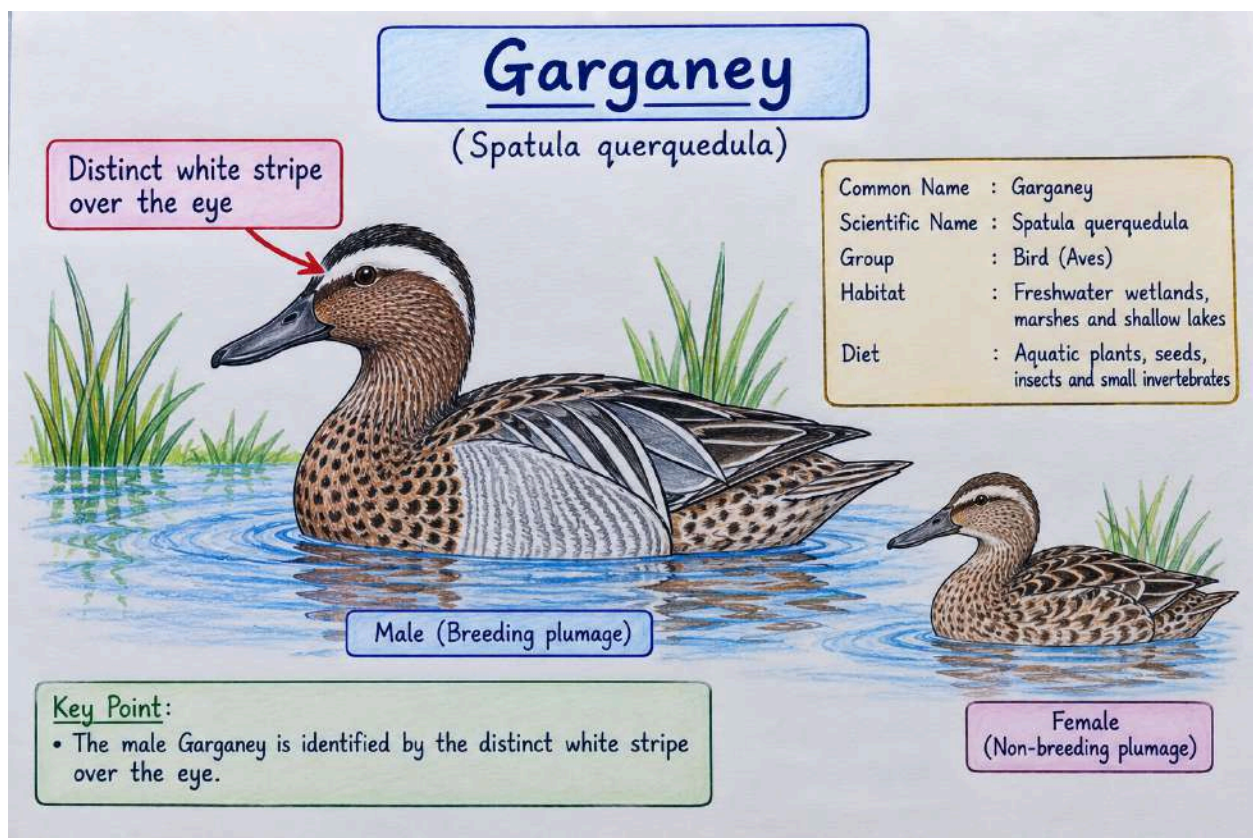
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2. Bar-headed Goose

- **Taxonomic Classification:**
 - Kingdom: Animalia
 - Phylum: Chordata

- Class: Aves
- Order: Anseriformes
- Scientific Name: *Anser indicus*
- **Migratory Status:** Winter Migrant (from Mongolia and Tibet).
- **Habitat in Maharashtra:** Large freshwater reservoirs and agricultural fields near water bodies (e.g., Jayakwadi Bird Sanctuary).
- **Characteristics:** Identifiable by two distinctive black bars on the back of its white head. It is one of the highest-flying birds in the world, capable of migrating directly over the Himalayas at altitudes exceeding 20,000 feet.
- **Primary Threats:** Poaching, agricultural poisoning (pesticides in fields where they forage), and degradation of freshwater lakes due to siltation.

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3. Garganey

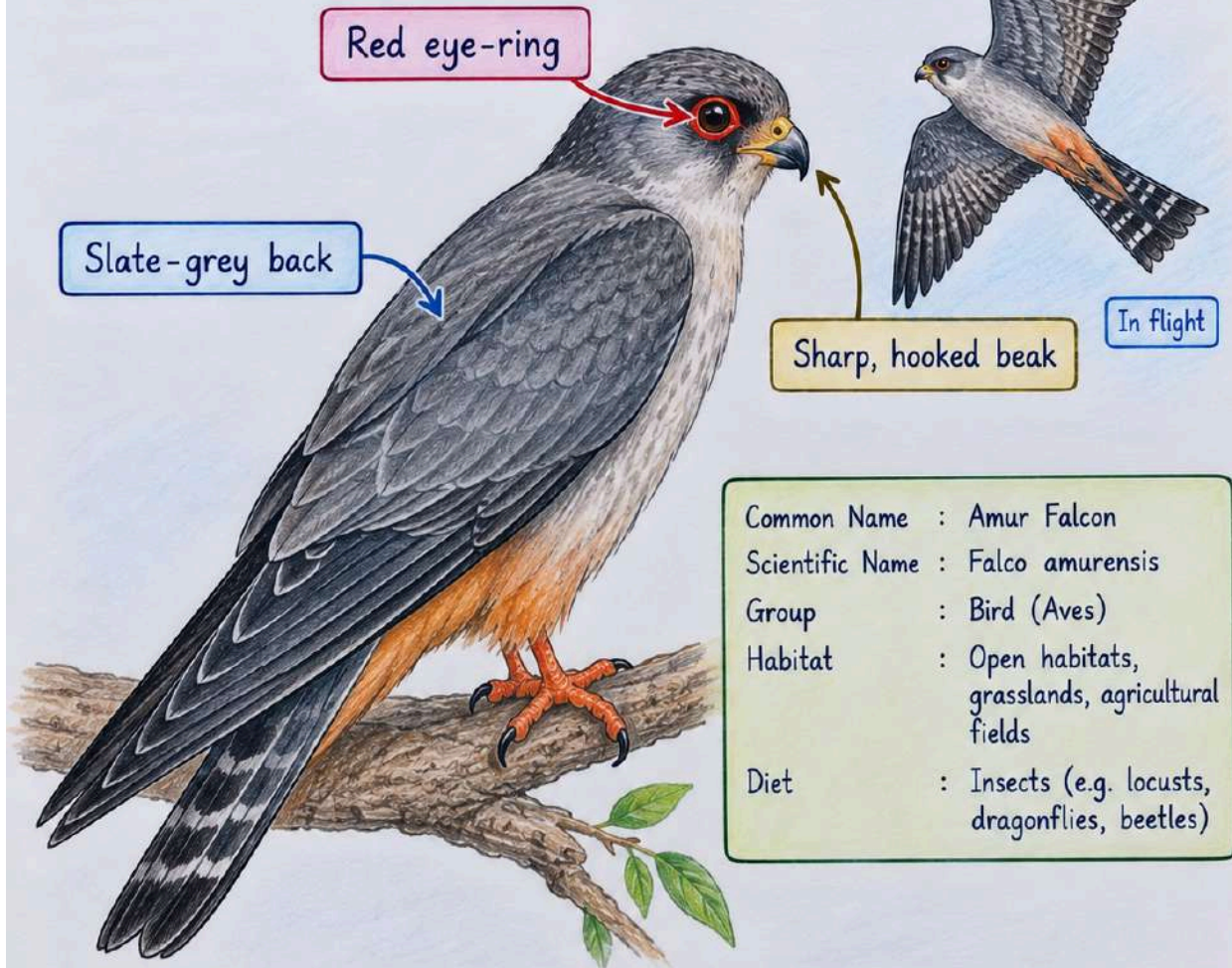
- **Taxonomic Classification:**
 - Kingdom: Animalia
 - Phylum: Chordata
 - Class: Aves
 - Order: Anseriformes
 - Scientific Name: *Spatula querquedula*

- **Migratory Status:** Winter Migrant (from Europe and Siberia).
- **Habitat in Maharashtra:** Shallow freshwater lakes, marshes, and flooded agricultural fields across the state.
- **Characteristics:** A small dabbling duck. The breeding male features a striking broad white stripe curving over the eye and extending down the neck. They travel in massive, highly synchronized flocks.
- **Primary Threats:** The draining of seasonal marshes for urban development and illegal trapping using nets in unprotected water bodies.

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Amur Falcon

(*Falco amurensis*)



Common Name	: Amur Falcon
Scientific Name	: <i>Falco amurensis</i>
Group	: Bird (Aves)
Habitat	: Open habitats, grasslands, agricultural fields
Diet	: Insects (e.g. locusts, dragonflies, beetles)

Key Points:

- Small falcon, known for its long-distance migration.
- Easily identified by its red eye-ring, slate-grey upperparts and sharp hooked beak.
- Winters in southern Africa, including India.

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Part II: Forest, Grassland, and Passage Migrants

4. Amur Falcon

- Taxonomic Classification:

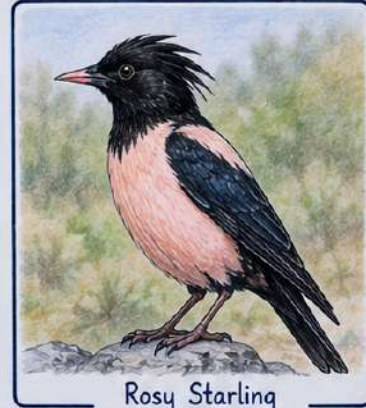
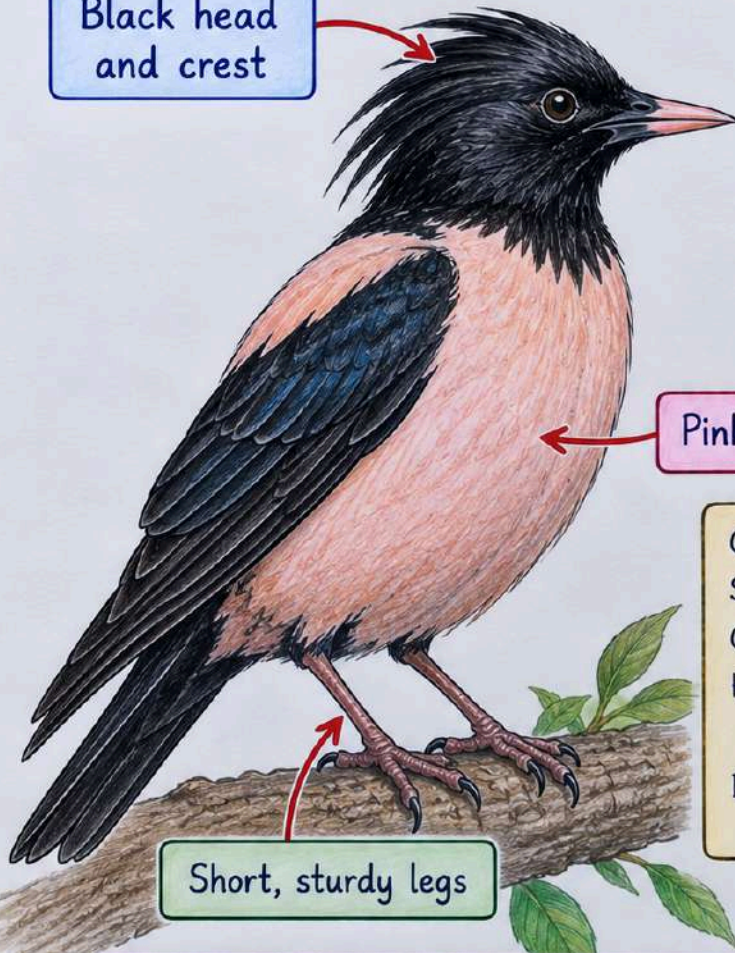
- Kingdom: Animalia
- Phylum: Chordata
- Class: Aves
- Order: Falconiformes
- Scientific Name: *Falco amurensis*
- **Migratory Status:** Passage Migrant (from Eastern Russia/China to South Africa).
- **Habitat in Maharashtra:** Open forests and hilly regions of the Western Ghats (frequently sighted resting in Lonavala and surrounding hills).
- **Characteristics:** A small, remarkable raptor (bird of prey) that undertakes one of the longest migratory journeys over the Arabian Sea. Males are slate-grey with distinct reddish-orange eye-rings and feet.
- **Primary Threats:** Habitat destruction in their resting corridors and historical hunting (though highly reduced now due to global conservation efforts).

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Rosy Starling

(Pastor roseus)

Black head
and crest



Rosy Starling
(in natural habitat)

Pink body

Short, sturdy legs

Common Name	: Rosy Starling
Scientific Name	: Pastor roseus
Group	: Bird (Aves)
Habitat	: Open country, farmlands, grasslands
Diet	: Insects, fruits, seeds, berries

Key Points:

- Distinctive pink body with black head and crest.
- Short, sturdy legs.
- Often seen in large flocks, especially during migration.
- Visits India in winter (a migratory bird).

5. Rosy Starling

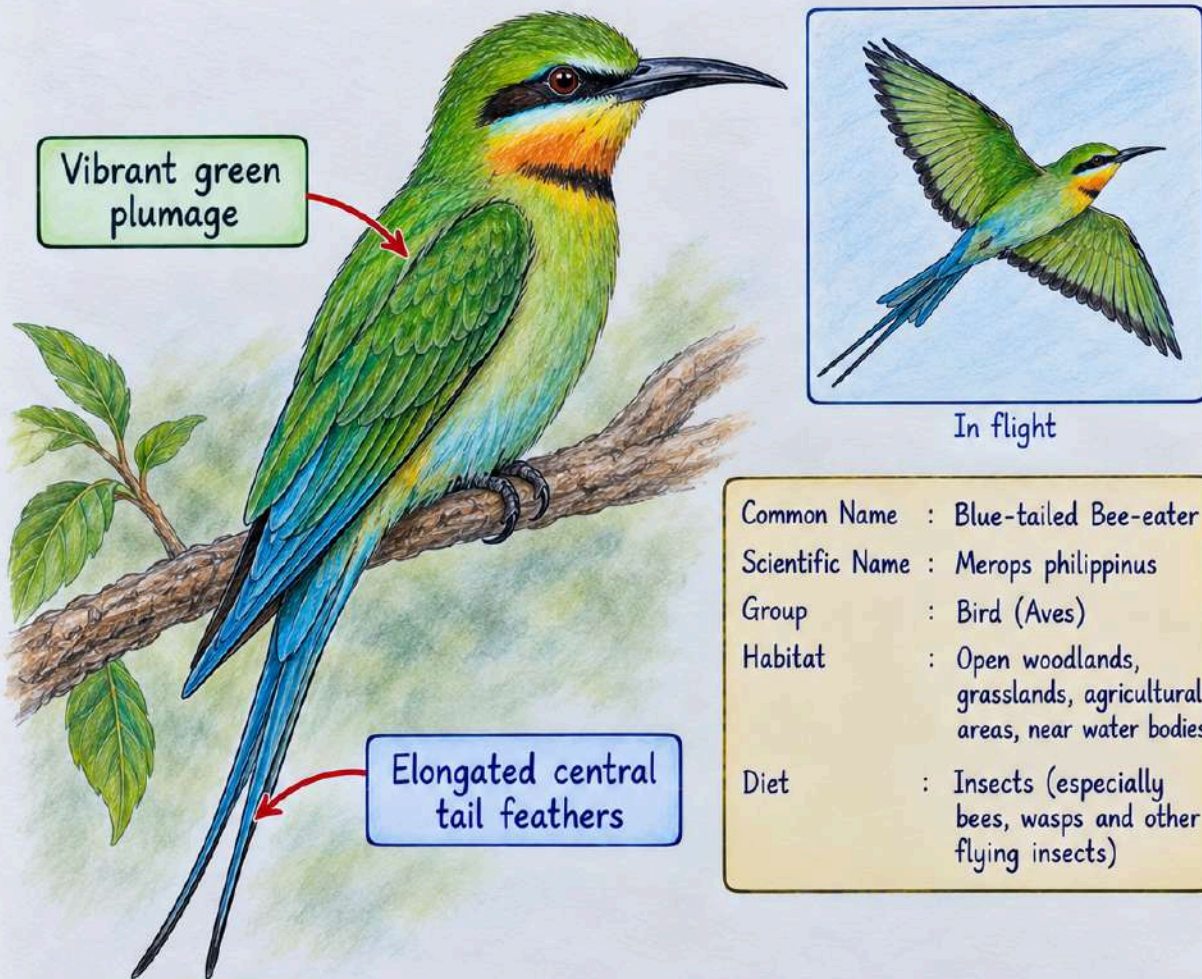
- **Taxonomic Classification:**
 - Kingdom: Animalia

- Phylum: Chordata
- Class: Aves
- Order: Passeriformes
- Scientific Name: *Pastor roseus*
- **Migratory Status:** Winter Migrant (from Eastern Europe and Central Asia).
- **Habitat in Maharashtra:** Agricultural lands, open grasslands, and urban scrublands throughout the Deccan Plateau.
- **Characteristics:** A highly gregarious bird featuring a beautiful pale pink body with a contrasting glossy black head, crest, wings, and tail. They move in massive, cloud-like swarms called murmurations.
- **Primary Threats:** Heavy usage of toxic chemical insecticides in agricultural fields, which depletes their primary food source (locusts and grasshoppers) and causes secondary poisoning.

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Blue-tailed Bee-eater

(*Merops philippinus*)



Common Name	: Blue-tailed Bee-eater
Scientific Name	: <i>Merops philippinus</i>
Group	: Bird (Aves)
Habitat	: Open woodlands, grasslands, agricultural areas, near water bodies
Diet	: Insects (especially bees, wasps and other flying insects)

Key Points :

- Noted for its vibrant green plumage and blue tail.
- Has elongated central tail feathers.
- Feeds mainly on flying insects like bees and wasps (hence the name "Bee-eater").
- Commonly seen in open areas, often perching on exposed branches or wires.
- A migratory bird, visiting India during the winter season.

6. Blue-tailed Bee-eater

- **Taxonomic Classification:**
 - Kingdom: Animalia

- Phylum: Chordata
- Class: Aves
- Order: Coraciiformes
- Scientific Name: *Merops philippinus*
- **Migratory Status:** Local/Summer Migrant (moving regionally within the subcontinent).
- **Habitat in Maharashtra:** Open woodlands, riverbanks, and coastal scrubs.
- **Characteristics:** A striking, richly colored bird with bright green plumage, a blue tail, and a black eye-stripe. It catches bees, dragonflies, and wasps in mid-air.
- **Primary Threats:** Destruction of sandy riverbanks where they burrow their nests, often caused by illegal sand mining activities.

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Demoiselle Crane

(*Grus virgo*)

White ear tufts

Long black
neck feathers



Close-up of head (side view)



Common Name	: Demoiselle Crane
Scientific Name	: <i>Grus virgo</i>
Group	: Bird (Aves)
Habitat	: Wetlands, marshes, grasslands, shallow lakes (especially in Central Asia, India and nearby regions)
Diet	: Grasses, aquatic plants, seeds, insects, small invertebrates

Key Points:

- White ear tufts are a key identification feature.
- Long black neck feathers give it a graceful look.
- Tall, slender bird with grey body and black head and neck.
- Common in wetlands and grasslands, especially during winter in India.

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7. Demoiselle Crane

- **Taxonomic Classification:**
 - Kingdom: Animalia
 - Phylum: Chordata
 - Class: Aves
 - Order: Gruiformes
 - Scientific Name: *Grus virgo*
- **Migratory Status:** Winter Migrant (from Eurasia).

- **Habitat in Maharashtra:** Dry grasslands and edges of shallow lakes.
- **Characteristics:** The smallest species of crane, recognized by a tuft of white feathers extending from behind the eye and long black feathers on the chest. They have a loud, trumpeting call.
- **Primary Threats:** Loss of wintering grassland habitats to urbanization and collisions with high-tension power lines.

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Summary of Migratory Birds in Maharashtra

Species Name	Scientific Name	Type	Habitat in MH
Greater Flamingo	<i>Phoenicopterus roseus</i>	Winter	Coastal Mudflats
Bar-headed Goose	<i>Anser indicus</i>	Winter	Freshwater Lakes
Garganey	<i>Spatula querquedula</i>	Winter	Shallow Marshes
Amur Falcon	<i>Falco amurensis</i>	Passage	Western Ghats
Rosy Starling	<i>Pastor roseus</i>	Winter	Agricultural lands
Blue-tailed Bee-eater	<i>Merops philippinus</i>	Summer	Riverbanks
Demoiselle Crane	<i>Grus virgo</i>	Winter	Grasslands

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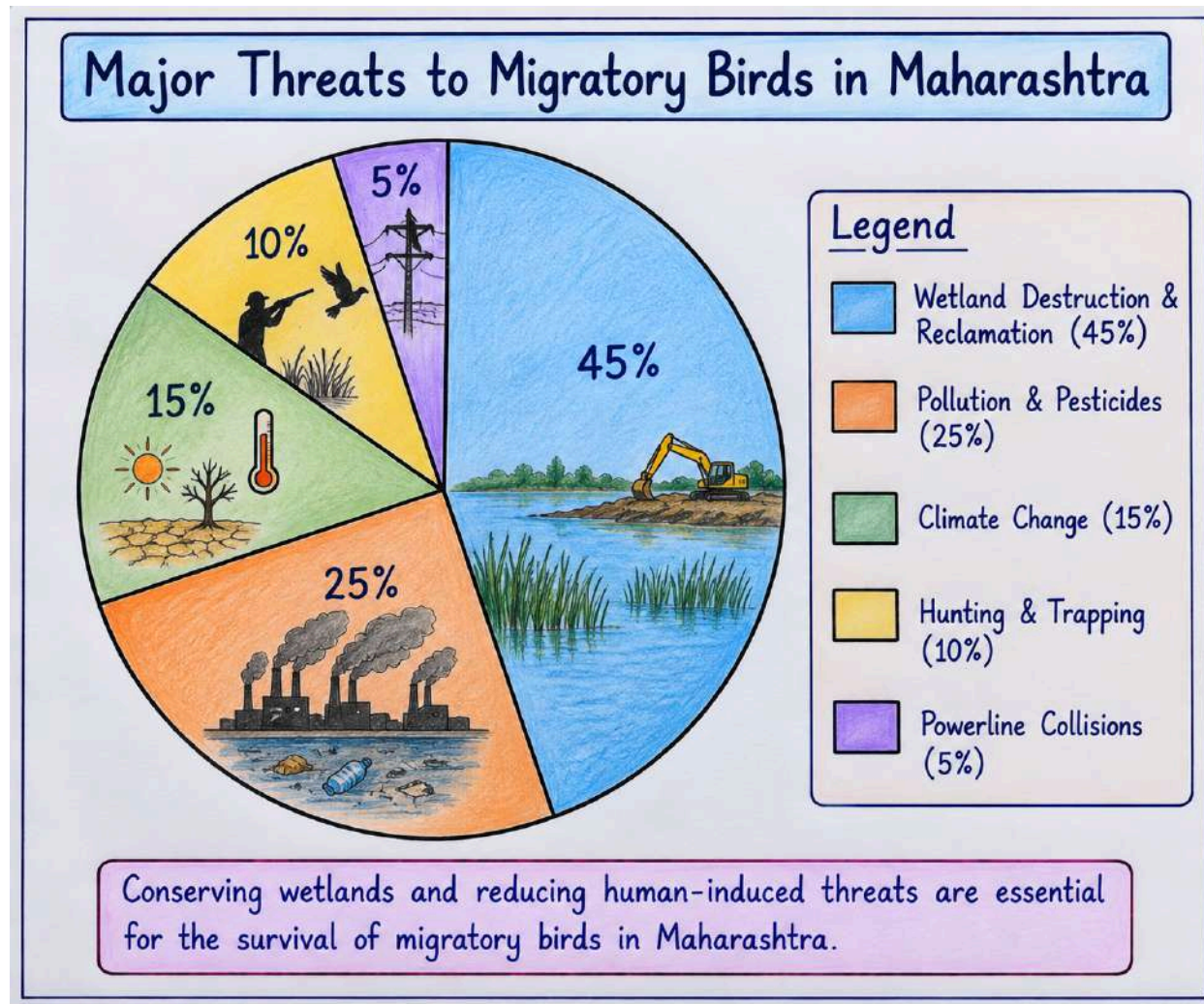
d. Analysis of collected data

Through the formulation of the gathered information, a clear pattern regarding the reliance of migratory birds on Maharashtra's ecosystems emerges.

1. Habitat Specificity and Dependency: The data definitively proves that different migratory species depend on highly specific micro-habitats. Flamingos require the unique salinity and mud texture of coastal creeks to filter-feed, while Bar-headed geese require expansive freshwater reservoirs. If these specific habitats are altered, the birds cannot simply relocate to a different environment; they perish.

2. The Danger of Habitat Loss (Anthropogenic Stress): The analysis reveals that human (anthropogenic) interference is the overwhelming threat to these migrations. Coastal reclamation in Mumbai for infrastructure destroys ancient mudflats.

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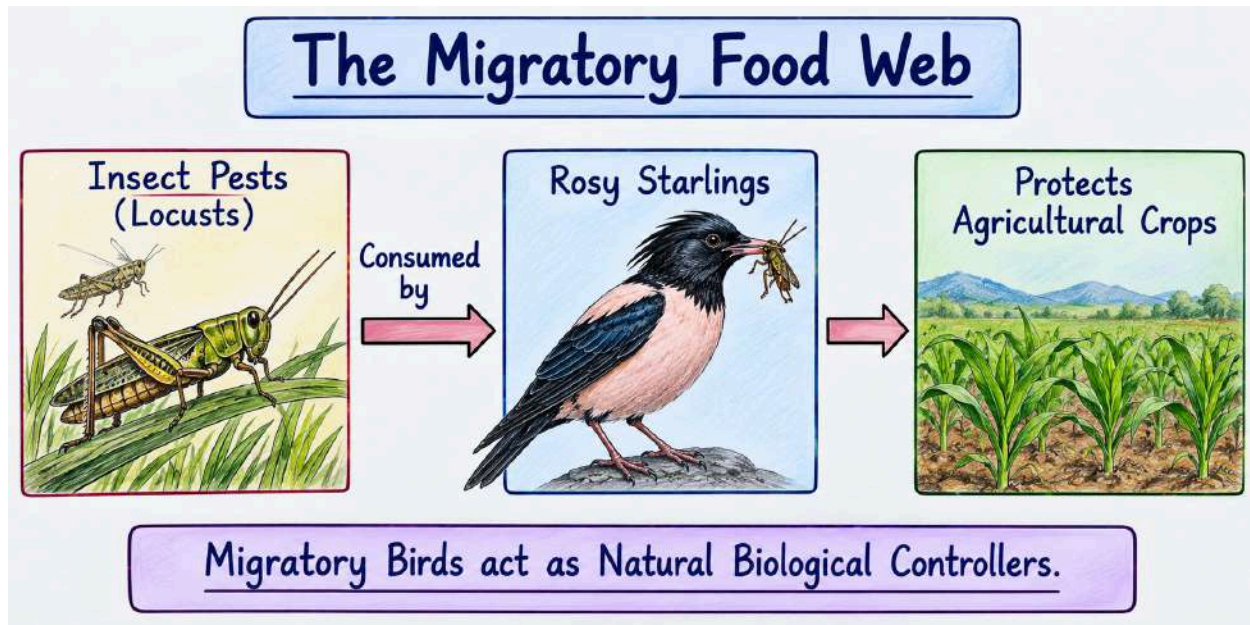
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Similarly, the draining of seasonal marshes and unregulated sand mining on riverbanks directly destroy the resting and feeding grounds that these birds have utilized for millennia.

3. Ecological Role of Migratory Birds: The compiled data highlights that these birds are not just visitors; they provide vital ecosystem services. Rosy Starlings act as highly effective biological pest controllers, consuming millions of crop-destroying locusts in agricultural fields. Waterfowls like the Garganey aid in the cross-pollination of aquatic plants and the dispersal of seeds across different water bodies, maintaining genetic diversity in aquatic flora.

4. The Threat of Pollution: Heavy usage of toxic chemical insecticides in agriculture leads to bioaccumulation in birds. Furthermore, industrial effluents dumped into coastal wetlands destroy the benthic fauna (micro-organisms) that wading birds rely on for sustenance.

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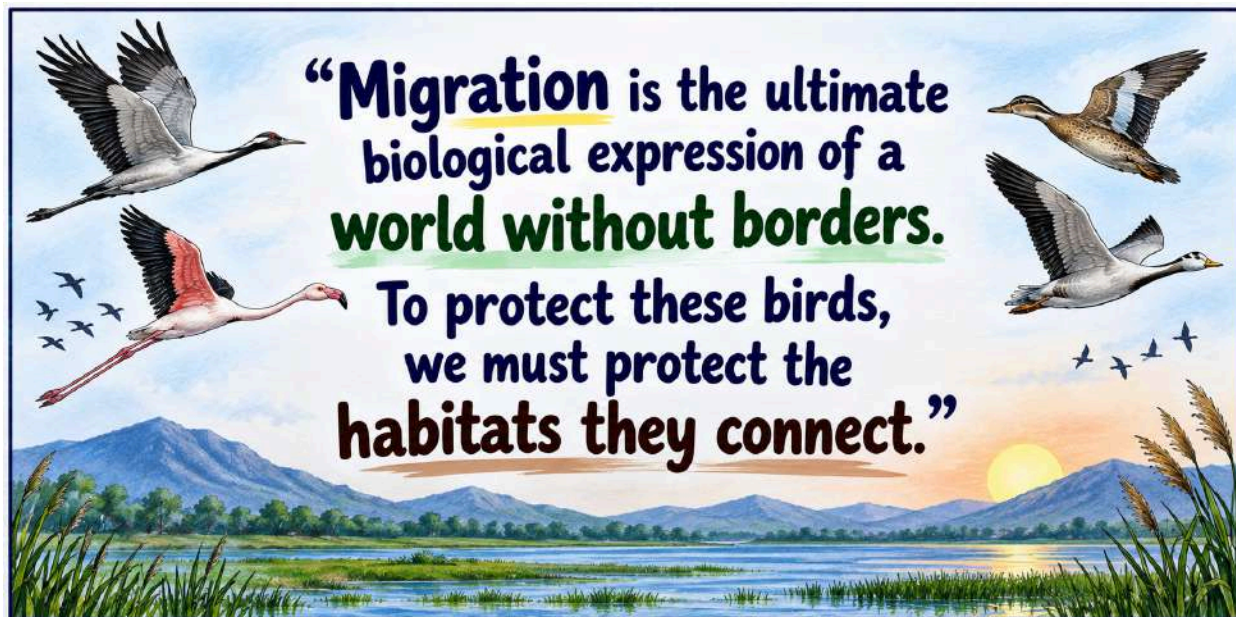
Conservation Strategies Based on Analysis: To secure the future of avian migration in Maharashtra, targeted conservation strategies must be amplified:

- **Protection of Wetlands:** Stricter enforcement of wetland conservation rules to prevent the draining and reclamation of critical habitats like Sewri, Ujani, and Jayakwadi.
- **Sustainable Agriculture:** Promoting organic farming practices to reduce pesticide runoff, which poisons both the birds and their food sources.
- **Mitigating Infrastructure Impact:** Installing bird diverters on high-tension power lines passing through known migratory corridors and grasslands to prevent fatal collisions.

e. Conclusion Arriving at a conclusion based on the analysis of data, it is evident that Maharashtra serves as an indispensable sanctuary for millions of migratory birds. This project successfully fulfills its primary objective of identifying key migratory species and their required habitats.

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(Write a bold, prominent quote in the center of the page: "Migration is the ultimate biological expression of a world without borders. To protect these birds, we must protect the habitats they connect.").



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The data conclusively proves that these avian migrations are under immense anthropogenic stress, primarily driven by rapid urbanization, wetland reclamation, and chemical pollution. The loss of these migratory routes is not just a tragedy for ornithology; it results in the loss of critical ecosystem services like natural pest control and seed dispersal, which directly impacts human agriculture.

Effective conservation requires recognizing wetlands and grasslands not as "wastelands" to be developed, but as vital ecological assets. By establishing community-led conservation reserves, promoting eco-tourism, and enforcing strict environmental protection laws, we can ensure that the skies of Maharashtra remain a welcoming haven for these incredible global travelers for generations to come.

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f. Bibliography- references for all the information sighted in the project

Textbooks and Academic Resources:

1. State Board Biology Textbook for Class 12, Maharashtra State Bureau of Textbook Production and Curriculum Research, Pune. (Referenced for core concepts of Ecosystems and Biodiversity).
2. NCERT Biology Class 12 Textbook.

Government & Scientific Reports:

1. "Birds of Maharashtra" – Field guides and publications by the Bombay Natural History Society (BNHS).

2. Asian Waterbird Census (AWC) Reports – Published by Wetlands International.
5. "State of India's Birds" Report – Collaborative research document assessing the conservation status of migratory species.

Online Scientific Databases:

1. eBird Database (ebird.org/india) - Accessed for confirming localized habitat sightings and migratory routes within Maharashtra.
2. The IUCN Red List of Threatened Species Database (iucnredlist.org).